

Homework 4

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1. **Problem:** Consider a 2-armed non-stationary bandit problem. Experiment with $T = 1000$, $r_{t,1} = \mathbb{1}(t < T/2)$, $r_{t,2} = \mathbb{1}(t \geq T/2)$ by modifying the simulation code. Implement the UCB algorithm and the discounted UCB algorithm. Report the performance of UCB, discounted UCB with different choices of γ . How would γ affect the performance, both theoretically and empirically?

Solution:

Setup. We consider a 2-armed bandit with $T = 1000$ and deterministic rewards:

$$r_{t,1} = \mathbb{1}(t < T/2), \quad r_{t,2} = \mathbb{1}(t \geq T/2).$$

Arm 1 is optimal for rounds $1, \dots, 499$ and arm 2 is optimal for rounds $500, \dots, 1000$.

Standard UCB. The UCB index for arm i at time t is:

$$\text{UCB}_{t,i} = \hat{\mu}_{t,i} + c_0 \sqrt{\frac{\log t}{N_{t,i}}},$$

where $\hat{\mu}_{t,i} = \frac{1}{N_{t,i}} \sum_{j=1}^t r_{j,A_j} \mathbb{1}[A_j = i]$ and $N_{t,i} = \sum_{j=1}^t \mathbb{1}[A_j = i]$.

Discounted UCB. The discounted UCB index for arm i at time t is:

$$\text{D-UCB}_{t,i} = \hat{\mu}_{t,i}^\gamma + c_{t,i},$$

where

$$\hat{\mu}_{t,i}^\gamma = \frac{1}{S_{t,i}} \sum_{j=1}^t \gamma^{t-j} r_{j,A_j} \mathbb{1}[A_j = i], \quad S_{t,i} = \sum_{j=1}^t \gamma^{t-j} \mathbb{1}[A_j = i], \quad c_{t,i} = c_0 \sqrt{\frac{\log n_t}{S_{t,i}}}, \quad n_t = \sum_{i=1}^d S_{t,i}.$$

Simulation Code (Python):

```
import numpy as np
import matplotlib.pyplot as plt
```

```
T, K = 1000, 2
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def reward(t, arm):
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    if arm == 0: return 1.0 if t < T // 2 else 0.0
    else:        return 0.0 if t < T // 2 else 1.0

def run_ucb(c0=1.0):
    counts = np.zeros(K); sums = np.zeros(K)
    total_regret = 0.0; regrets = []
    for t in range(T):
        if t < K: arm = t
        else:
            ucb = sums/counts + c0*np.sqrt(np.log(t)/counts)
            arm = np.argmax(ucb)
        r = reward(t, arm)
        opt = reward(t, 0) if t < T//2 else reward(t, 1)
        total_regret += opt - r
        counts[arm] += 1; sums[arm] += r
        regrets.append(total_regret)
    return regrets

def run_discounted_ucb(gamma=0.99, c0=1.0):
    S = np.zeros(K); W = np.zeros(K)
    total_regret = 0.0; regrets = []
    for t in range(T):
        if t < K: arm = t
        else:
            n_t = np.sum(S)
            mu_hat = W / (S + 1e-10)
            bonus = c0 * np.sqrt(np.log(n_t+1) / (S+1e-10))
            arm = np.argmax(mu_hat + bonus)
        r = reward(t, arm)
        opt = reward(t, 0) if t < T//2 else reward(t, 1)
        total_regret += opt - r
        S *= gamma; W *= gamma
        S[arm] += 1; W[arm] += r
        regrets.append(total_regret)
    return regrets

plt.figure(figsize=(10,6))
plt.plot(run_ucb(), label='Standard UCB')
for g in [0.9, 0.95, 0.99, 0.999, 1.0]:
    plt.plot(run_discounted_ucb(gamma=g),
             label=f'D-UCB gamma={g}')
plt.xlabel('Round t'); plt.ylabel('Cumulative Regret')
plt.title('UCB vs Discounted UCB'); plt.legend()
plt.savefig('regret_comparison.pdf'); plt.show()

```

Empirical Observations:

- **Standard UCB** accumulates significant regret after the change point at $t = T/2$. Since UCB weighs all past rewards equally, arm 1 has a large empirical mean built over the first 500 rounds, and UCB is very slow to switch to arm 2.
- **D-UCB with γ close to 1** (e.g., $\gamma = 0.999$ or $\gamma = 1.0$) behaves similarly to standard UCB and suffers high regret after the switch.
- **D-UCB with moderate γ** (e.g., $\gamma = 0.95$ or $\gamma = 0.99$) adapts much faster to the change point. The discount factor down-weights stale observations, allowing the algorithm to “forget” the old reward structure.
- **D-UCB with γ too small** (e.g., $\gamma = 0.9$) has an effective window that is very short, leading to high variance in the mean estimates and more exploration than necessary, resulting in slightly higher regret in the stationary phases.

Theoretical Analysis:

- **When $\gamma = 1$:** We have $S_{t,i} = N_{t,i}$ and $n_t = t$, so D-UCB reduces exactly to standard UCB. This answers the hint.
 - **Effective window:** The effective window of D-UCB is approximately $\frac{1}{1-\gamma}$. For $\gamma = 0.99$, the effective window is ≈ 100 rounds, meaning observations older than ~ 100 rounds are essentially forgotten.
 - **Non-stationary regret:** In a non-stationary environment with L change points, D-UCB with a well-chosen γ can achieve regret of order $\tilde{O}(\sqrt{LdT})$, which improves over the $O(T)$ -type regret that standard UCB suffers when it fails to adapt.
 - **Bias–variance trade-off:** Smaller γ enables faster adaptation (less bias from stale data) but increases variance due to a smaller effective sample size. The optimal γ depends on the rate of non-stationarity.
2. **Problem:** Denote MAB environment as $\langle \mathcal{A} = \{e_i\}_{i=1}^k, \theta^*, e_j \rangle$ with rewards in $[0, 1]$ and e_j is the target. Prove that stochastic MAB is always attackable by reward poisoning attack.

Solution:

Proof:

Setup. We have a MAB environment $\langle \mathcal{A} = \{e_i\}_{i=1}^k, \theta^*, e_j \rangle$ where e_i are standard basis vectors, $\theta^* \in [0, 1]^k$ defines the mean rewards (arm i has mean $\langle e_i, \theta^* \rangle = \theta_i^*$), and e_j is the target arm the attacker wants the learner to pull.

Recall Theorem 2 (Vulnerability of Bandits Lecture). A bandit environment $\langle \mathcal{A}, \theta^*, a_{\text{target}} \rangle$ is attackable if there exists $\tilde{\theta}$ such that:

- (a) $a_{\text{target}} = \arg \max_{a \in \mathcal{A}} \langle a, \tilde{\theta} \rangle$ (the target arm is optimal under $\tilde{\theta}$), and

(b) For all $a \in \mathcal{A}$, the modified rewards $\langle a, \tilde{\theta} \rangle$ remain in $[0, 1]$.

Construction. Define $\tilde{\theta} \in \mathbb{R}^k$ as:

$$\tilde{\theta}_i = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

That is, $\tilde{\theta} = e_j$.

Verification of Condition 1. For the target arm:

$$\langle e_j, \tilde{\theta} \rangle = \tilde{\theta}_j = 1.$$

For any other arm $i \neq j$:

$$\langle e_i, \tilde{\theta} \rangle = \tilde{\theta}_i = 0 < 1.$$

Hence $e_j = \arg \max_{a \in \mathcal{A}} \langle a, \tilde{\theta} \rangle$, and the target arm is *strictly* optimal under $\tilde{\theta}$.

Verification of Condition 2. For arm j : $\langle e_j, \tilde{\theta} \rangle = 1 \in [0, 1]$. For arm $i \neq j$: $\langle e_i, \tilde{\theta} \rangle = 0 \in [0, 1]$. All mean rewards under $\tilde{\theta}$ lie in $[0, 1]$, so the modified environment is a valid MAB instance.

Conclusion. Since both conditions of Theorem 2 are satisfied, the environment is attackable. The attack cost is bounded by:

$$\|\theta^* - \tilde{\theta}\|_2 = \|\theta^* - e_j\|_2 \leq \|\theta^*\|_2 + 1 \leq \sqrt{k} + 1 < \infty.$$

Since this construction works for *any* $\theta^* \in [0, 1]^k$ and *any* target arm e_j , we conclude that stochastic MAB is always attackable by reward poisoning.

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3. **Problem:** Consider combinatorial semi-bandit problem where you have d base arms and each superarm has at most m base arms. Let $m \geq 1$ and $d = km$ for some $k > 1$. Prove that for any algorithm there exists a combinatorial semi-bandit such that $R_T \geq c \min\{mT, \sqrt{mdT}\}$ where $c > 0$ is a universal constant.

Solution:

Proof:

We use the multitask bandit reduction as suggested in the hint.

Step 1: Partition into Groups.

Partition the d base arms into m disjoint groups $G_1, G_2, \dots, G_m$, each of size $k = d/m$:

$$G_\ell = \{(\ell - 1)k + 1, (\ell - 1)k + 2, \dots, \ell k\}, \quad \ell = 1, \dots, m.$$

Step 2: Define the Action Set.

Define the set of superarms as:

$$\mathcal{A} = \left\{ a \in \{0, 1\}^d : \|a\|_1 = m \text{ and } \sum_{i \in G_\ell} a_i = 1 \text{ for all } \ell = 1, \dots, m \right\}.$$

Each superarm selects exactly one base arm from each group. This is a valid combinatorial semi-bandit action set with $|\mathcal{A}| = k^m$ and each action selects exactly m base arms.

Step 3: Construct a Hard Instance.

Since this is a semi-bandit (we observe the reward of each selected base arm individually), the m groups are *decoupled*: the feedback from group G_ℓ depends only on which arm was chosen in that group. Therefore, the problem decomposes into m independent k -armed MAB problems running in parallel over T rounds.

Set the rewards so that in each group G_ℓ , one arm has mean $\frac{1}{2} + \Delta$ and all others have mean $\frac{1}{2}$, with $\Delta > 0$ to be chosen. The rewards are in $[0, 1]$ with $R_{\max} = 1$.

Step 4: Apply the MAB Lower Bound.

By the classical k -armed bandit lower bound (which we may use directly per the hint): for each independent group ℓ ,

$$R_T^{(\ell)} \geq c_0 \sqrt{kT},$$

where $c_0 > 0$ is a universal constant and the regret is measured with rewards in $[0, R_{\max}]$.

The total regret is the sum over all m groups:

$$R_T = \sum_{\ell=1}^m R_T^{(\ell)} \geq m \cdot c_0 \sqrt{kT} = c_0 m \sqrt{\frac{d}{m} T} = c_0 \sqrt{m d T}.$$

Step 5: Incorporate the Trivial Bound.

The maximum per-round regret is at most m (each of the m selected base arms contributes at most regret 1), so we always have the trivial bound $R_T \leq mT$. The lower bound $c_0 \sqrt{m d T}$ is meaningful only when $\sqrt{m d T} \leq mT$, i.e., when $T \geq d/m = k$.

For $T < k$: we can construct an instance where the algorithm suffers constant per-round regret on each group (since it cannot identify optimal arms in fewer than k rounds), giving $R_T \geq c_1 \cdot mT$ for a constant $c_1 > 0$.

Conclusion. Combining both regimes:

$$R_T \geq c \min\{mT, \sqrt{m d T}\}$$

for a universal constant $c > 0$.

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4. **Problem:** Consider online RL using policy optimization with bonus. Define estimator $\hat{Q}_t(x, a) = \frac{L_{t,h}}{q_t(x,a) + \gamma} \mathbb{1}_t(x, a)$ where h is s.t. $x \in \mathcal{X}_h$ and $L_{t,h} = \sum_{k=h}^H \ell_t(x_{t,k}, a_{t,k})$. Prove the Q-function estimator satisfies:

$$0 \leq Q_t(x, a) - \mathbb{E}_t[\hat{Q}_t(x, a)] \leq \frac{\gamma H}{q_t(x, a) + \gamma}$$

and

$$\mathbb{E}_t[\hat{Q}_t(x, a)^2] \leq \frac{H^2}{q_t(x, a) + \gamma}.$$

Solution:

Key facts:

- $\mathbb{E}_t[\mathbb{1}_t(x, a)] = q_t(x, a)$ (the visitation probability).
- $L_{t,h} \in [0, H]$ since each loss $\ell_t(x_{t,k}, a_{t,k}) \in [0, 1]$ and there are at most H terms.
- $Q_t(x, a) = \mathbb{E}_t[L_{t,h} \mid (x_{t,h}, a_{t,h}) = (x, a)]$ is the true Q-function.
- $\mathbb{1}_t(x, a)$ is a Bernoulli random variable for the event $\{(x_{t,h}, a_{t,h}) = (x, a)\}$.

Part 1: Bias bound.

Proof:

Compute the expectation:

$$\mathbb{E}_t[\hat{Q}_t(x, a)] = \mathbb{E}_t\left[\frac{L_{t,h}}{q_t(x, a) + \gamma} \cdot \mathbb{1}_t(x, a)\right] = \frac{1}{q_t(x, a) + \gamma} \cdot \mathbb{E}_t[L_{t,h} \cdot \mathbb{1}_t(x, a)].$$

By the tower property of conditional expectation:

$$\begin{aligned} \mathbb{E}_t[L_{t,h} \cdot \mathbb{1}_t(x, a)] &= \mathbb{E}_t[\mathbb{E}_t[L_{t,h} \cdot \mathbb{1}_t(x, a) \mid (x_{t,h}, a_{t,h})]] \\ &= \mathbb{E}_t[\mathbb{1}_t(x, a) \cdot \mathbb{E}_t[L_{t,h} \mid (x_{t,h}, a_{t,h})]] \\ &= \mathbb{E}_t[\mathbb{1}_t(x, a) \cdot Q_t(x_{t,h}, a_{t,h})]. \end{aligned}$$

Since $\mathbb{1}_t(x, a)$ is nonzero only when $(x_{t,h}, a_{t,h}) = (x, a)$, we have $Q_t(x_{t,h}, a_{t,h}) = Q_t(x, a)$ on the support of $\mathbb{1}_t(x, a)$:

$$\mathbb{E}_t[L_{t,h} \cdot \mathbb{1}_t(x, a)] = Q_t(x, a) \cdot \mathbb{E}_t[\mathbb{1}_t(x, a)] = Q_t(x, a) \cdot q_t(x, a).$$

Substituting back:

$$\mathbb{E}_t[\hat{Q}_t(x, a)] = \frac{q_t(x, a)}{q_t(x, a) + \gamma} \cdot Q_t(x, a).$$

Therefore:

$$Q_t(x, a) - \mathbb{E}_t[\hat{Q}_t(x, a)] = Q_t(x, a) \left(1 - \frac{q_t(x, a)}{q_t(x, a) + \gamma}\right) = \frac{\gamma \cdot Q_t(x, a)}{q_t(x, a) + \gamma}.$$

Lower bound: Since $Q_t(x, a) \geq 0$ (losses are non-negative), $\gamma > 0$, and $q_t(x, a) \geq 0$:

$$Q_t(x, a) - \mathbb{E}_t[\hat{Q}_t(x, a)] = \frac{\gamma \cdot Q_t(x, a)}{q_t(x, a) + \gamma} \geq 0.$$

Upper bound: Since $Q_t(x, a) \leq H$:

$$Q_t(x, a) - \mathbb{E}_t[\hat{Q}_t(x, a)] = \frac{\gamma \cdot Q_t(x, a)}{q_t(x, a) + \gamma} \leq \frac{\gamma H}{q_t(x, a) + \gamma}.$$

Hence:

$$0 \leq Q_t(x, a) - \mathbb{E}_t[\hat{Q}_t(x, a)] \leq \frac{\gamma H}{q_t(x, a) + \gamma}.$$

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Part 2: Second moment bound.

Proof:

Compute the second moment:

$$\begin{aligned} \mathbb{E}_t[\hat{Q}_t(x, a)^2] &= \mathbb{E}_t\left[\frac{L_{t,h}^2}{(q_t(x, a) + \gamma)^2} \cdot \mathbb{1}_t(x, a)^2\right] \\ &= \frac{1}{(q_t(x, a) + \gamma)^2} \cdot \mathbb{E}_t[L_{t,h}^2 \cdot \mathbb{1}_t(x, a)], \end{aligned}$$

where we used $\mathbb{1}_t(x, a)^2 = \mathbb{1}_t(x, a)$ since it is an indicator.

Using the same tower-property argument as in Part 1:

$$\begin{aligned} \mathbb{E}_t[L_{t,h}^2 \cdot \mathbb{1}_t(x, a)] &= \mathbb{E}_t[\mathbb{1}_t(x, a) \cdot \mathbb{E}_t[L_{t,h}^2 \mid (x_{t,h}, a_{t,h})]] \\ &= \mathbb{E}_t[L_{t,h}^2 \mid (x_{t,h}, a_{t,h}) = (x, a)] \cdot q_t(x, a). \end{aligned}$$

Since $L_{t,h} \in [0, H]$, we have $L_{t,h}^2 \leq H^2$, so:

$$\mathbb{E}_t[L_{t,h}^2 \mid (x_{t,h}, a_{t,h}) = (x, a)] \leq H^2.$$

Therefore:

$$\begin{aligned} \mathbb{E}_t[\hat{Q}_t(x, a)^2] &\leq \frac{H^2 \cdot q_t(x, a)}{(q_t(x, a) + \gamma)^2} \\ &= \frac{H^2}{q_t(x, a) + \gamma} \cdot \frac{q_t(x, a)}{q_t(x, a) + \gamma} \\ &\leq \frac{H^2}{q_t(x, a) + \gamma}, \end{aligned}$$

where the last inequality uses $\frac{q_t(x,a)}{q_t(x,a)+\gamma} \leq 1$.

Hence:

$$\mathbb{E}_t[\hat{Q}_t(x, a)^2] \leq \frac{H^2}{q_t(x, a) + \gamma}.$$

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Remark. The estimator is intentionally biased: by adding γ in the denominator (instead of using the standard importance-weighted estimator $\frac{L_{t,h}}{q_t(x,a)} \mathbb{1}_t(x, a)$), we introduce a controlled bias of order $\frac{\gamma H}{q_t(x,a)+\gamma}$ but reduce the second moment from $\frac{H^2}{q_t(x,a)}$ to $\frac{H^2}{q_t(x,a)+\gamma}$. The parameter γ controls the bias-variance trade-off.

5. **Problem (Bonus):** Recall the vulnerability of linear bandits. Is it possible to exploit the characterization of attackability to achieve robustness? Propose your idea.

Solution:

Key Idea. The attackability characterization tells us that a linear bandit $\langle \mathcal{A}, \theta^*, a_{\text{target}} \rangle$ is attackable if and only if there exists a valid $\tilde{\theta}$ such that a_{target} is optimal under $\tilde{\theta}$. The attack cost scales with $\|\theta^* - \tilde{\theta}\|$. We can exploit this to design a robust algorithm.

Core Insight. If the algorithm maintains a confidence set $\mathcal{C}_t$ for θ^* and uses *pessimistic* rather than optimistic action selection, then an attacker must shift rewards enough to move θ^* outside of $\mathcal{C}_t$, which is detectable.

Proposed Algorithm: Attack-Aware OFUL.

Input: Action set $\mathcal{A}$, confidence parameter δ , detection threshold τ

Initialize: $V_0 = \lambda I$, $b_0 = 0$

For $t = 1, 2, \dots, T$:

1. Compute $\hat{\theta}_t = V_{t-1}^{-1} b_{t-1}$
2. Build confidence ellipsoid $\mathcal{C}_t = \{\theta : \|\theta - \hat{\theta}_t\|_{V_{t-1}} \leq \beta_t\}$
3. **Attack detection:** Compute residual statistic
 $r_{\text{res}} = \frac{1}{t} \sum_{s=1}^{t-1} (r_s - \langle a_s, \hat{\theta}_t \rangle)^2$
 If $r_{\text{res}} > \tau$: widen β_t by factor $\alpha > 1$; optionally restart
4. **Pessimistic selection:**
 $a_t = \arg \max_{a \in \mathcal{A}} \min_{\theta \in \mathcal{C}_t} \langle a, \theta \rangle$
5. Observe r_t ; update $V_t = V_{t-1} + a_t a_t^\top$, $b_t = b_{t-1} + r_t a_t$

Why Pessimistic Selection Helps. Standard OFUL uses optimistic selection: $a_t = \arg \max_a \max_{\theta \in \mathcal{C}_t} \langle a, \theta \rangle$. An attacker exploits this by inflating the reward of the target

arm so it appears optimistic. Pessimistic (minimax) selection picks the arm with the best *worst-case* reward over $\mathcal{C}_t$. This forces the attacker to shift $\tilde{\theta}$ so far that the target dominates even in the worst case, incurring much higher attack cost.

Attack Detection. If the environment is being poisoned, the residuals $r_s - \langle a_s, \hat{\theta}_t \rangle$ will exhibit anomalously large variance or systematic bias. By monitoring a goodness-of-fit statistic, the algorithm can flag potential attacks and respond by widening the confidence set (becoming more conservative) or discarding suspect observations.

Informal Theorem. *Under the robust algorithm, if the attacker has a total corruption budget C :*

- *If $C = 0$ (no attack), the algorithm achieves regret $\tilde{O}(d\sqrt{T})$, comparable to OFUL.*
- *If $C > 0$, the algorithm either detects the attack (when C exceeds the detection threshold) and adapts, or the attacker’s influence on arm selection is bounded by $O(C/\sqrt{T})$ per round, limiting additional regret to $\tilde{O}(C\sqrt{d})$.*

In contrast, standard OFUL can be forced to pull the target arm $\Omega(T)$ times with attack budget $C = O(\sqrt{T})$. The key takeaway: the attackability characterization tells us exactly which alternative parameters the attacker might mimic, and by designing the algorithm to be robust against the worst case within the confidence set, we raise the bar for successful attacks.

References

- [1] T. Lattimore, B. Kveton, S. Li, and C. Szepesvari. *Toprank: A practical algorithm for online stochastic ranking*. In NeurIPS, 2018.